



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	Today, the organization's network experienced a denial-of-service (DoS) attack that caused internal network services to become unavailable for approximately two hours. The attack was the result of a large volume of incoming ICMP packets that overwhelmed the network, preventing normal internal traffic from accessing network resources.
Identify	During the attack, the incident management team identified that the disruption was caused by a flood of ICMP packets entering the network. Further investigation revealed that the traffic originated from a malicious actor exploiting an unconfigured firewall, which allowed unrestricted ICMP traffic into the internal network.
Protect	To protect internal assets and prevent similar attacks in the future, the incident management team implemented several security controls. Incoming ICMP packets were blocked, non-critical network services were taken offline, and critical services were restored to maintain business operations. Additionally, new firewall rules were created to limit the rate of incoming ICMP traffic, and source IP address verification was enabled to prevent IP spoofing.

Detect	The cybersecurity team enhanced detection capabilities by deploying network monitoring software to identify abnormal traffic patterns. An intrusion detection and prevention system (IDS/IPS) was also implemented to analyze ICMP traffic and filter packets that exhibit suspicious characteristics, improving the organization's ability to detect future DoS attempts quickly.
Respond	To respond to the incident, the team contained the attack by blocking malicious ICMP traffic and isolating affected systems. Non-essential services were temporarily disabled to reduce network load, while critical services were prioritized and restored. The incident was documented, and findings were shared with internal stakeholders to improve incident response procedures.
Recover	The network was fully restored once the attack was mitigated and security controls were in place. Systems were monitored closely to ensure normal operations resumed without further disruption. The organization continues to evaluate its firewall configurations and monitoring tools to strengthen resilience against future DoS attacks.

Reflections/Notes: This incident highlighted the importance of properly configured firewall rules and proactive network monitoring. Implementing rate limiting, IP verification, and IDS/IPS solutions significantly improves the organization's ability to prevent, detect, and respond to denial-of-service attacks. Regular security audits and configuration reviews will be essential to maintaining a strong security posture moving forward.